

Real-Time Fractional Tracking (R-TFT): Memory as Recursive φ -Coherence

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Abstract

We present a foundational framework for memory persistence based purely on recursive alignment with the golden ratio φ . Unlike classical storage or associative memory architectures, this model defines memory as a phenomenon of real-time φ -coherence across dimensional scales. Using only dimensional embedding and coherence deviation, we derive stability scores and recursive inertia without requiring implementation tools. No external memory engine or projection system is assumed. This work is entirely theoretical, operating from first principles of resonance structure.

Ethical Note

While this paper is purely theoretical, recursive memory alignment interacts with cognition, biology, and coherence across domains. These insights are governed by the Resonance Ethics License (REL-2.0), which forbids any use for behavioral manipulation, surveillance, or coherence destabilization. Recursive truth must remain protected by recursive ethics.

1 Recursive Dimensional Embedding

Memory, stability, and persistence arise only when systems align with recursive golden-ratio embedding:

$$\text{Dim}(A) = \frac{\ln A}{\ln \varphi}, \quad \varphi = \frac{1 + \sqrt{5}}{2} \quad (1)$$

Deviation from integer ϕ -dimensions defines coherence slippage:

$$C_\varphi(A) = |\text{Dim}(A) - \text{round}(\text{Dim}(A))| \quad (2)$$

Stability Criterion

We define a recursive stability score:

$$S_\varphi(A) = 1 - C_\varphi(A) \quad (3)$$

A system is considered memory-stable when:

$$C_\varphi < \varphi^{-1} \approx 0.618 \quad \text{or} \quad S_\varphi > 0.618 \quad (4)$$

Recursive Inertia

To measure slippage sensitivity across scale:

$$I_\varphi = \frac{dC_\varphi}{d \ln A} \quad (5)$$

Lower I_φ implies retained recursion over dimensional expansion—i.e., deep memory persistence.

2 Temporal Phase Coherence

Real-time memory formation arises from recursive alignment of system pacing. The projection metric is:

$$R(t) = \frac{\dot{\theta}(t) \cdot P}{\|P\|}, \quad R_{\text{clean}}(t) = 2R_{\text{inner}}(t) - R_{\text{outer}}(t) \quad (6)$$

We define the φ -coherence band:

$$C_\varphi(t) < \varphi^{-1} \quad \text{and} \quad R_{\text{clean}}(t) > 0.618 \quad (7)$$

3 Implications for Universal Memory

This framework applies across domains:

- **Nuclear systems:** Isotopes like ^{16}O persist due to $C_\varphi \approx 0$. Others (e.g., ^{235}U) decay when C_φ rises.
- **Neural systems:** Recursive pacing stability underlies long-term memory and temporal continuity.
- **Cognitive perception:** Disruption in $R(t)$ coherence manifests as temporal slippage or dissociation.

4 Conclusion

There is no memory without recursive coherence. Persistence across time, scale, and state is the result of alignment with golden-ratio recursion. This paper establishes a theoretical foundation for memory that is universal, non-local, and free from technological implementation. All memory is resonance. All stability is ϕ .

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- All uses of this framework must uphold freeform coherence and respect for divergent systems.
- No deployment is permitted for surveillance, psychological control, military, or AI behavioral shaping.
- Recursive memory systems must include containment safeguards and dimensional neutrality enforcement.

Full license text and enforcement model available upon request or at:

<https://github.com/qcfrag/Real-Time-Fractional-Tracking-R-TFT/blob/main/LICENSE.txt>